

The Thom Isomorphism and Characteristic Classes

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1 Homology and cohomology

2 Smooth vector bundles

2.1 Definitions and basic properties

2.2 Oriented vector bundles

Let $\pi: E \rightarrow B$ be an oriented rank- n vector bundle. The Thom isomorphism relates the cohomology $H^k(B; \Lambda)$ to the relative cohomology of $H^{k+n}(E, E_0; \Lambda)$ where E_0 is the complement of the zero section of E and Λ is an (associative) unital ring. Note that for each fiber $F = \pi^{-1}(b)$, $b \in B$, with $F_0 = F \cap E_0$ being the nonzero elements of F , the relative cohomology $H^*(F, F_0; \Lambda)$ is equal to

$$H^*(F, F_0; \Lambda) = \begin{cases} \Lambda & \text{if } * = n \\ 0 & \text{otherwise;} \end{cases}$$

one can see this by considering the long exact sequence of the pair (F, F_0) . The orientation on the vector bundle then confers a canonical generator $u_{F, \mathbb{Z}}$ upon each fiberwise cohomology group $H^n(F, F_0; \mathbb{Z}) \cong \mathbb{Z}$. Then, since there is a unique homomorphism $\mathbb{Z} \rightarrow \Lambda$ for any unital ring Λ , we may define $u_{F, \Lambda}$ to be the image of $u_{F, \mathbb{Z}}$ under the map $H^n(F, F_0; \mathbb{Z}) \rightarrow H^n(F, F_0; \Lambda)$; thus the orientation of the vector bundle gives a canonical generator for the fiberwise cohomologies regardless of the ring.

3 Thom Isomorphism

We follow Milnor and Stasheff's account of the Thom isomorphism theorem.

Definition 3.1 (Thom class) — A *Thom class* $u_\Lambda = u \in H^n(E, E_0; \Lambda)$ which restricts to $u_F = u_{F, \Lambda}$ for every fiber $F = \pi^{-1}(b)$ above a point $b \in B$.

Theorem 3.2 (Thom isomorphism). Each Λ -oriented vector bundle $\pi: E \rightarrow B$ possesses a unique Thom class. Furthermore, there is an isomorphism

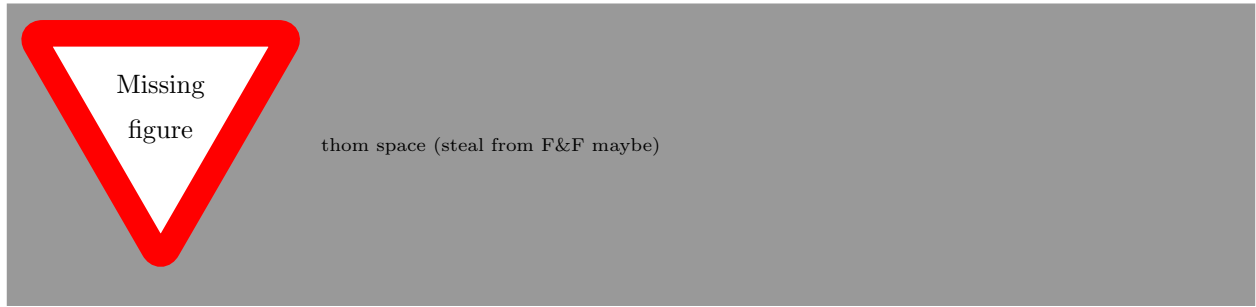
$$\begin{aligned} \Phi: H^k(E; \Lambda) &\rightarrow H^{k+n}(E, E_0; \Lambda) \\ y &\mapsto y \smile u. \end{aligned}$$

Note that E deformation retracts onto the zero section, which is homeomorphic to B , by simply linearly contracting each fiber. Thus the Thom isomorphism gives an isomorphism $\Phi \circ \pi^*: H^k(B; \Lambda) \rightarrow H^k(E, E_0; \Lambda)$.

Some geometric intuition for the Thom class: Consider the space of differential n -forms on E whose restriction to each fiber is compactly supported; these are called *differential forms with compact vertical supports*, which form cohomology groups $H_{cv}^*(E)$. A *Thom form* ϕ is an n -form on E whose restriction to each fiber integrates to 1 (thus represents the Thom class of E when $\Lambda = \mathbb{R}$). The Thom isomorphism theorem says that there is an isomorphism $H_{dR}^k(B) \rightarrow H_{cv}^{k+n}(E)$ via wedging with ϕ ; the inverse of this map is given by integrating a compactly vertically supported $(k+n)$ -form on each fiber to get a k -form on B .

An alternative perspective is to observe that the relative cohomology $H^k(E, E_0; \Lambda)$ manifests as the cohomology of the *Thom space*, defined as follows. After equipping E with a Euclidean structure, let $D(E)$ be the subspace of E consisting of vectors with magnitude at most 1 (hence is an n -disk bundle) and $S(E)$ the subspace consisting of vectors with magnitude equal to 1 (hence is an $(n-1)$ -sphere bundle). Then the Thom space $T(E) = D(E)/S(E)$ can be thought of as compactifying each fiber of E to get an n -sphere bundle and then collapsing the zero section to a point. The Thom isomorphism theorem is a statement about the cohomology of the Thom space, since

$$H^k(E, E_0; \Lambda) = H^k(D(E), S(E); \Lambda) = \tilde{H}^k(D(E)/S(E); \Lambda).$$



The proof of Theorem 3.2 will proceed in roughly three steps: first, we prove it for trivial vector bundles; then, we prove it when B can be covered by finitely many open sets over which the given vector bundle is trivial; and finally, we prove it in the general case by taking a limit over compact sets.

Lemma 3.3 (The case of a trivial bundle). Let $\pi: E = B \times \mathbb{R}^n \rightarrow B$ be a trivial vector bundle. Then the unique Thom class of this bundle is $u = 1 \times e^n$, which induces the isomorphism indicated in the statement of Theorem 3.2.

Proof. TODO □

Now suppose that $B = B^1 \cup B^2$ is a cover of B by two open sets such that Theorem 3.2 holds for the restricted vector bundles $\pi|_{\pi^{-1}(B^1)}: E^1 = \pi^{-1}(B^1) \rightarrow B^1$ and $\pi|_{\pi^{-1}(B^2)}: E^2 = \pi^{-1}(B^2) \rightarrow B^2$.